Redistributing Transportation Costs to Resolve Inequalities

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[PRELIMINARY. PLEASE DO NOT CIRCULATE.]

1. Motivation: Poverty Correlated w/ Supply Costs Across Space

- When poverty is correlated with supply costs across space, firms may charge the poor higher prices
- To resolve such inequality, the government, for example, can provide transportation subsidies to firms
    - [1] How might such government programs be designed?
    - [2] What issues arise in the government's interaction with firms?
    - [3] And what are the welfare effects of monitoring firms?
- We study these issues by examining Ghana's Unified Petroleum Pricing Fund Program – the only scalable program we know of in the world that redistributes transportation costs to resolve inequalities

Paper Overview: The Unified Petroleum Pricing Fund Program

- Very large budget-neutral redistribution program targeting transportation costs:
  [1] Reimburse distribution firms of petroleum products for their transport costs (in shipping products from O/storage-depots to D/retail-outlets)
  [2] Ensures equal prices for a given firm across space (but not across firms)
  [2] Each firm sets their price (spatially uniform) for a period of two weeks, then revises it after every two-week window...
  [3] Poor live farther from supply, generating spatial inequalities [social protection?]
  [4] Spatial inequalities in petroleum prices pose security concerns [national security?]
  [5] High-transport cost and low-government capacity environment [poor monitoring?]

- Document trouble in achieving direct redistribution, arising from firm misconduct (noncompliance/fraudulent claims/misreporting)
- Eval. potential of using monitoring technologies to improve capacity and Program
- Present a Framework – to describe the Program, understand firm decisions, quantify welfare and counterfactuals for policy experimentation

Results Overview

- Transport costs are large (eq. 7% of tot gov. expenditures), much larger than total budget for "all" traditional social protection programs
- Significant firm misconduct, leading to 12% loss in revenues under the Fund
- Monitoring technologies eliminate this misconduct and/or leakages (-12%)
  - No effects on firm exits
  - Supply to distant locations increased, to further resolve spatial inequalities
- But firms increased their prices (+15%) due to the monitoring

[1] Reveals trade-off between incentives for firm misconduct and consumer welfare

[2] The optimal policy design would minimize this empirically observed trade-off between incentives for firm misconduct and consumer welfare under monitoring

[3] Results illustrate why firm misconduct (incl. corruption in government programs) entrenched in developing countries

Contributions to the Literature

- Transport costs [Teravaninthorn-Raballand (2009), Atkin-Donaldson (2015)]
  - Don't document only how large transport costs are but present a practical approach to redistributing such costs
- Social protection [Banerjee et al. (2024)]: First to highlight redistribution of transport costs, which we emphasize is a significant source of inequality
  - Design of program we present does not require identifying specific beneficiaries, and so, not limited by the large informal sector in developing countries
- Digital governance: Look at 'firm behavior' and their responses to monitoring technologies, not government officials (bureaucrats) or households, to improve the public redistribution program
  - Growing interest in improving government programs with digital technology [Muralidharan et al. (2016); Dodge et al. (2021); Banerjee et al. (2023)]
  - Yet all focused on households, despite importance of government-to-firm interactions

Outline

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2. Setting: Program Redistributing Transportation Costs and Uniform Pricing to Resolve Inequalities

- Views about the Program: mainly, to redistribute transport costs?
- Program details

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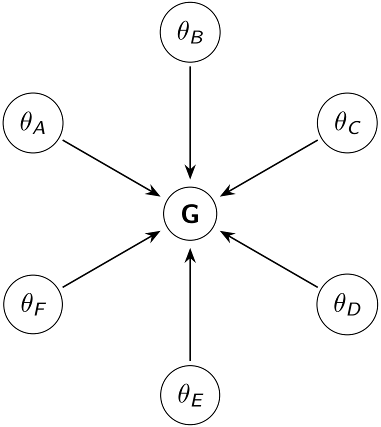
3. Model: Explaining the Program and Optimal Redistribution



- Distribution Network Topology: Hub and Spoke
- **Hub:** global market/farther away depot vs **Spoke:** local market/nearby depot
- Firms with quantity O=1 to sell (supply inelastically to whole of Ghana because they are importers), choose share **v** (volume) to move to **G** (global market)
- Firms differ in transportation costs (distance) per unit of transported volume to global markets θ with density f

Markets and Firms

- Local demand and market clearing

$$Q^l(p_l) = Q_l p_l^{-\varepsilon_{Dl}}, \qquad S - v = Q^l(p_l)$$

- That is the equilibrium local price $p_l = (\frac{Q_l}{S-v})^{1/\varepsilon_{Dl}}$
- Firm with transportation cost θ choose local monopoly price $p_l(\theta)$ but global price determined by global demand and market clearing

$$Q^G(p) = Q_G p^{-\varepsilon_{DG}}, \qquad \int v(\theta) df(\theta) = Q^G(p)$$

- Firms maximize profit by choosing p_l and share volume $\mathbf{v}$ they move to global market

$$\Pi = \max_{v, p_l} p_l(S - v) + v(p - \theta)$$

- After local price-setting, firms maximize:

$$\max_{v} Q_l^{1/\varepsilon_{Dl}}(S - v)^{1-1/\varepsilon_{Dl}} + v(p - \theta)$$

Markets and Firms

- We get assuming $\varepsilon_{DI} > 1$: If $\theta > p$ then v=0 and firms don't wanna send anything and profit is $\Pi = Q_I^{1/\varepsilon_{DI}}$
- If $\theta \leq$ p, then v according to, $(1 - 1/\varepsilon_{DI}) Q_I^{1/\varepsilon_{DI}} (S - v)^{-1/\varepsilon_{DI}} = p - \theta$

- In summary: firms with high transportation cost won't want to send oil to the global market.
- So, if the government puts important weight on this global market, it will want to introduce subsidies **[so, rationalizing the Program]**.
- Note that firms do not internalize own shipping decision on price in global market (price takers), while a planner would do.

Redistributing Transportation Costs: Program

- Now let's set a tax, a uniform oil tax τ (margin or price build-up) and subsidy per unit of moved volume t(θ) (transport cost reimbursement)

- Firms maximize profit $\max_{v,p_l} p_l(1-\tau)(S-v) + v(p(1-\tau) + t(\theta) - \theta)$
- After local price set,

$$\max_v Q_l^{1/\varepsilon_{Dl}}(S-v)^{1-1/\varepsilon_{Dl}}(1-\tau) + v(p(1-\tau) + t(\theta) - \theta)$$

- We get assuming $\varepsilon > 1$: If $\theta > p(1-\tau) + t(\theta)$ then v=0 and don't wanna send anything and profit is $\Pi = Q_l^{1/\varepsilon_{Dl}}(1-\tau)$
- If $\theta \leq p(1-\tau) + t(\theta)$, then v according to,

$$1 - 1/\varepsilon_{Dl})Q_l(S - v(\theta))^{-1/\varepsilon_{Dl}}(1-\tau) = p(1-\tau) + t(\theta) - \theta$$

- Example: linear subsidy with slope t, $t(\theta) = t\theta$ enables more firms $\theta \in [0, \tilde{\theta}]$ to move volume to global market, i.e

$$\tilde{\theta} = \frac{p(1-\tau)}{1-t} \tag{1}$$

The Optimal Reimbursement Schedule

- Government's problem [for the Program] maximizes weighted consumer surplus

$$\max_{t(\theta),\tau} \mu^G \int_p^\infty Q^G(p')dp' + \mu^L \int \int_{p_l(\theta)} Q^L(p')dp'df(\theta) \tag{2}$$

s. to

$$v(\theta)df(\theta) = Q^G(p) = Q_G p^{-\varepsilon_{D_G}}$$
$$Q^L(p_l(\theta)) = Q_L p_l(\theta)^{-\varepsilon_{Dl}} = S - v(\theta) \text{ [local demand]}$$
$$(1 - 1/\varepsilon_{Dl})Q_l(S - v(\theta))^{-1/\varepsilon_{Dl}} = p + \frac{t(\theta) - \theta}{1 - \tau} \text{ for } \theta \leq p(1-\tau) + t(\theta)\text{: FOC}_v$$
$$\int \tau[p_l(\theta)(S - v(\theta)) + pv(\theta)]df(\theta) = \int t(\theta)v(\theta)df(\theta) \text{ [gov. budget constraint]}$$

- Optimize and derive sufficient statistics for optimal subsidy schedule
- Survey government to estimate welfare weights (μ^G, μ^L)
- Compare to empirical reimbursement schedule implemented by the government.

Monitoring and Firm Misconduct

- Can overreport a share of volume $\hat{v} = (1 + u)v$ and pays a reputational cost ξ per misreported share
- Monitoring technology with probability of being caught $h(u)$
- Subsidy paid on reported volume $\hat{v}$ is not caught, and on real volume v if caught.
- Firm's problem to choose shipped volume v, reported volume $\hat{v}$ and monopolistic local market price p_l

$$\max_{v,\hat{v},p_l} [p_l(1-\tau)(S-v) + vp((1-\tau)-\theta) + t(\theta)[(1-h(u))\hat{v} + h(u)v] - \xi u] \tag{3}$$

- So if isoelastic function for example $h(u) = \delta \frac{u^\eta}{\eta}$ then

$$u^*(v;\theta) = \left(\frac{1-\xi/t(\theta)}{\delta v} \frac{\eta}{1+\eta} \right)^{1/\eta} \tag{4}$$

- Optimal misreporting share u^* decreasing in ξ, δ, and volume v, but increasing in the subsidy $t(\theta)$

Government's Problem

- Government's problem maximize weighted consumer surplus

$$\max_{t(\theta),\tau_g} \left[\mu^G \int_p^\infty Q^G(p')\, dp' + \mu^L \int \int_{p_l(\theta)} Q^L(p')\, dp'\, df(\theta) \right] \tag{5}$$

subject to:

$$\int v(\theta) df(\theta) = Q^G(p) = Q_G p^{-\varepsilon_{D_G}} \quad \text{(global market clearing)} \tag{6}$$

$$Q^L(p_l(\theta)) = Q_L p_l(\theta)^{-\varepsilon_{Dl}} = S(\theta) - v(\theta) \quad \text{(local demand)} \tag{7}$$

$$v \in \arg\max_v \left[Q_l^{1/\varepsilon_{Dl}} (S - v)^{1-1/\varepsilon_{Dl}} (1-\tau) + v(p(1-\tau) + t(\theta) - \theta) + t(\theta)[(1 - \delta h(u^*))u^* v - \xi u^* \right] \tag{8}$$

$$u^* = \left(\frac{1 - \xi / t(\theta)}{\delta v} \frac{\eta}{1+\eta} \right)^{1/\eta} \quad \text{(misreporting)} \tag{9}$$

$$\int \tau \left[p_l(\theta)(S(\theta) - v(\theta)) + pv(\theta) \right] df(\theta) = \int t(\theta) v(\theta) df(\theta) + \int t(\theta)(1 - \delta h(u^*))u^* v \quad \text{(gov bc)}$$

(10)

Estimation

The model takes as input $\{\Theta, S(\Theta), Q_l, Q_d, \varepsilon^{Dl}, \varepsilon^{Dg}, \tau, \theta_0, t, \xi, \delta, \eta\}$

- Empirical distribution of distance
- Empirical distribution of total firm supply per θ
- t is the empirical slope of the regression of the subsidies
- ξ, δ, η Endogenously estimated to match the elastictiy of our leakage and price

The model generates $\{p_l(\Theta), v(\Theta), u(\Theta), p, t(\Theta), v(\Theta)t(\Theta)\}$ And allows to calculate:

- Consumer surplus in status quo
- Producer surplus in status quo
- Optimal subsidy schedule and tax t, τ
- Optimal monitoring technology δ

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4. Data & Measurements

- **Survey** of government officials and distribution firms
  [1] Views about the Program
  [2] Field questions to measure welfare weights (μ^G, μ^L)
  [3] Forecasts about effects of monitoring technology BRVT

- **Administrative data** of universe of all shipments nationwide and Program records
  [1] We partner with the government [National Petroleum Authority NPA]
  [2] Digitize archival records at the NPA
  [3] Universe of all shipments nationwide
  [4] Spans 2011-2023, at specific order x date x truck x firm-level
    - Variable Descriptions
    - Summary Statistics

  [5] +Global oil prices from Energy Information Administration EIA [www.eia.gov]

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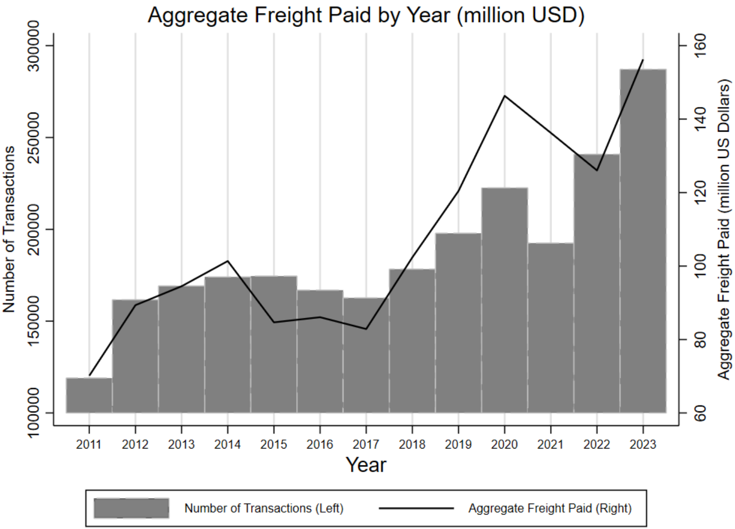
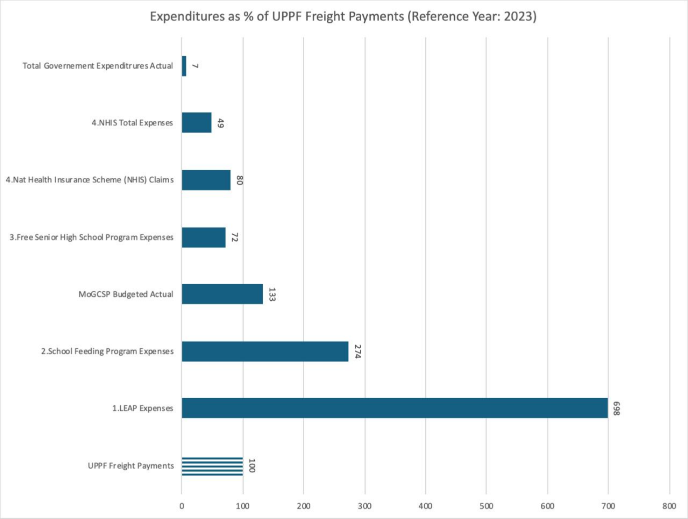
5. Results: Very Large Program

- Transport cost reimbursements: avg. of US$100 mil/yr & 200,000 distributions/yr



Very Large Program

- eq. 7% of gov expenditure, 133% of Social Protection Ministry's budget
- Re-imagine scope of social protection, to include non-traditional programs



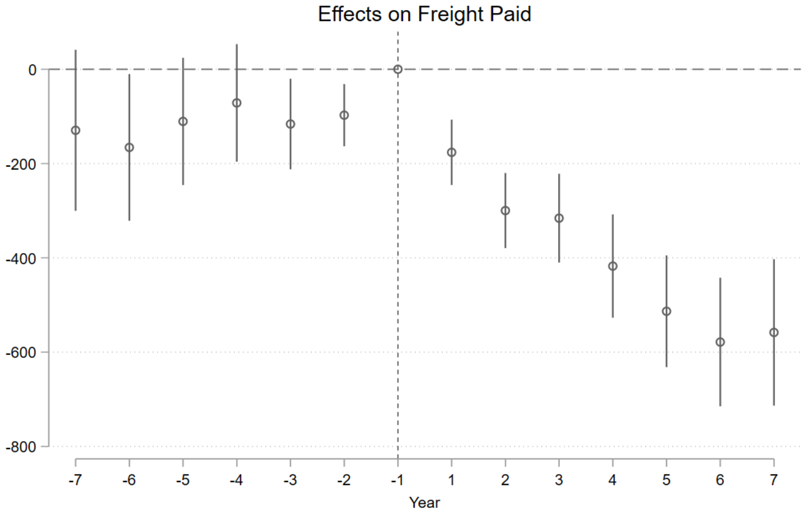
Regressions - Empirical Strategy

Truck (BRV)-Level Analysis:

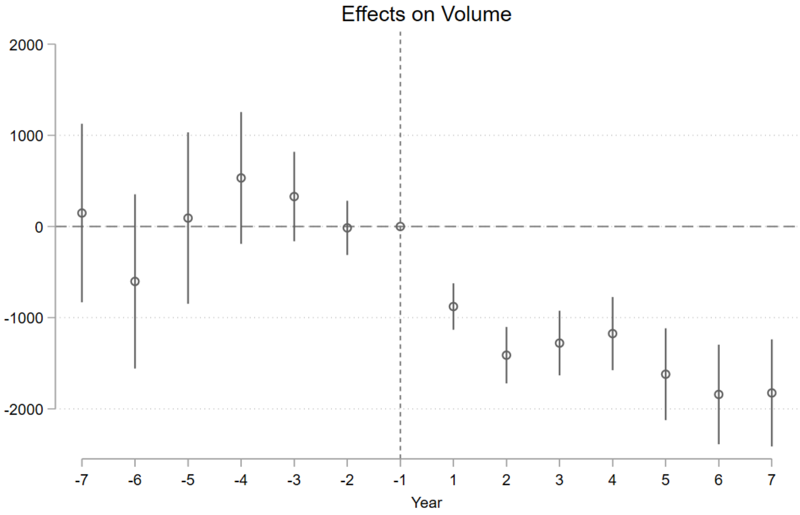
$$y_{ikODt} = \alpha_0 + \beta * \textbf{Treat}_k * post_t + \gamma * \textbf{Treat}_k + \alpha_O + \alpha_D + \delta_t + g_r(t) + \epsilon_{ikODt}$$

- y_{ikODt} outcome measures for order i by BRV truck k from origin O to destination D on date t
- $\textbf{Treat}_k$ equals 1 if truck k was ever tracked
- $post_t$ equals 1 if order was delivered after installation of trackers
- FEs: origin point O, destination D, order date t
- Region- or district-specific linear time trends: $g_r(t)$
- Clustered standard errors at firm- or distr company level

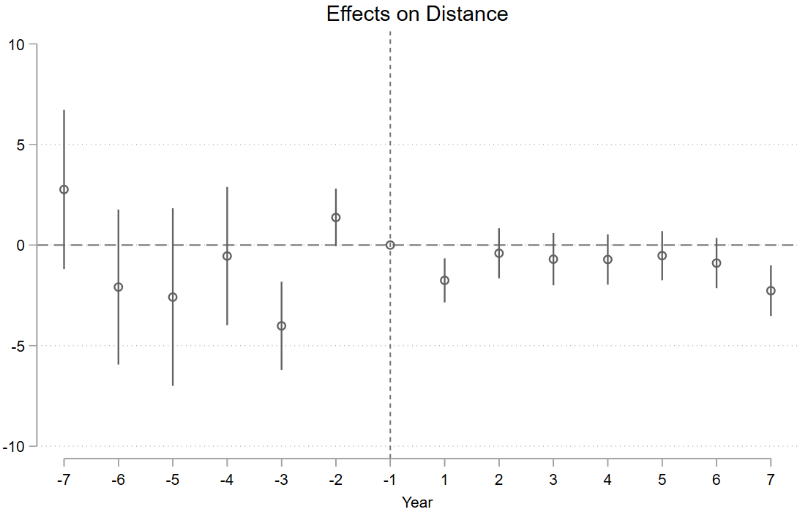
Effect of BRVT Monitoring on Program Outcome: Freight Paid



Effect of BRVT on Volume



Effect of BRVT on Distance



Effect of BRVT Monitoring Technology on Program Outcomes

Table 1: Effects of Monitoring Technology, BRVT, on Program Outcomes

VARIABLES	Freight Paid: Claims (GHS)	Volume (L)	Distance (Km)	Distance (Km)
Treat	-29.327	-298.024	-0.287	1.279
	(36.083)	(235.674)	(0.754)	(1.944)
x post	-293.417***	-1,374.591***	-0.633	8.213***
	(37.182)	(202.335)	(0.545)	(1.921)
Origin FE	Yes	Yes	Yes	
Destination FE	Yes	Yes	Yes	
Date FE	Yes	Yes	Yes	Yes
Region x Linear Trend	Yes	Yes	Yes	Yes
R-squared	0.508	0.110	0.898	0.533
Mean DV	3071.4	17430.0	158.2	158.2
Observations	2,440,419	2,440,420	2,373,135	2,379,575

Notes: Table regresses outcome measures for freight claim records. All columns include origin point, destination, and order date fixed effects. Region- or district-linear time trends are included. Standard errors are clustered at distribution company level and shown in parentheses.
∗∗∗Significant at the 1 percent level.
∗∗Significant at the 5 percent level.
∗Significant at the 10 percent level.

We include the period from year 2011 to 2023.

Effect of BRVT Monitoring Technology on Firm Prices

Firm (OMC)-Level Analysis:

$$y_{ct} = \alpha_0 + \beta * \textbf{SHARE}_{ct} + \alpha_c + \delta_t + g_c(t) + \epsilon_{ct}$$

- y_{ct} is outcome measures for firm c on time t
- $\textbf{SHARE}_{ct}$ equals # linked-BRV trucks tracked divided by # BRV trucks registered
- FEs: distribution firm c, year t
- Firm-specific linear time trends: $g_c(t)$
- Clustered standard errors at firm- or distr company level

Effect of BRVT Monitoring Technology on Firm Prices

Table 2: Effects of Monitoring Technology, BRVT, on Firms Pricing Decisions

VARIABLES	Price (GHS/L)		
	PMS: Petrol	AGO: Diesel	White Products (Petrol+Diesel)
SHARE	91.001*	107.329***	99.461**
	(49.571)	(37.255)	(42.250)
Year FE	Yes	Yes	Yes
OMC FE	Yes	Yes	Yes
Fuel Type FE			Yes
OMC x Linear Trend	Yes	Yes	Yes
R-squared	0.495	0.598	0.544
Mean DV	600.1	626.8	613.3
Observations	8,128	8,105	16,244

Notes: PMS denotes Premium Motor Spirit or petrol. AGO denotes Automotive Gas Oil or diesel. All columns include firm, and year fixed effects. Firm-linear time trends are included. Standard errors are clustered at distribution company level and shown in parentheses.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

We include the period from year 2016 to 2022.

Recap

- New evidence from Program that redistribute transport costs to resolve inequalities in practice
    - [1] Document the trouble in achieving direct redistribution
    - [2] Arises from firm misconduct (noncompliance/fraudulent claims/ misreporting)
- Monitoring technologies improve Program but undermined by firm price responses
    - [1] Firm misconduct decreased (eq. 12% of revenues)
    - [2] Firms increased prices (eq. around 15%)
- Several heterogeneity analysis, consistent w/ main results (in the 'future' paper?)
- Results generate design lessons for the Program & welfare

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6. Quantitative Analysis & Experimentation

- Combine empirical results and model
- Quantify welfare [the difference between no-monitoring and monitoring counterfactuals]
- Empirical trade-off between reduced misconduct (increased welfare) vs increased prices (decreased welfare)
- Use model to simulate 'no-uppf' counterfactual, where firms set different prices across space and quantify welfare

- Adjusting reimbursement subsidy schedule
- Increasing firm competition (spatially? decreasing p* at global market G)

- Recover firm distribution: homogeneous vs heterogeneous w/o monitoring
- So, why didn't distribution firms set higher prices pre-monitoring? signals as cheating? ask in firm surveys?